

Problem

Sketch the Bode plots for

$$\mathbf{H}(\omega) = \frac{0.2(10 + j\omega)}{j\omega(2 + j\omega)}$$

Step-by-step solution

Step 1 of 9

Consider the following transfer function:

$$\mathbf{H}(\omega) = \frac{0.2(10 + j\omega)}{j\omega(2 + j\omega)}$$

Simplify the transfer function.

$$\begin{aligned}\mathbf{H}(\omega) &= \frac{(0.2)(10)\left(1 + \frac{j\omega}{10}\right)}{(2)j\omega\left(1 + \frac{j\omega}{2}\right)} \\ &= \frac{\left(1 + \frac{j\omega}{10}\right)}{j\omega\left(1 + \frac{j\omega}{2}\right)}\end{aligned}$$

$$\mathbf{H}(\omega) = \frac{\left|1 + \frac{j\omega}{10}\right|}{|j\omega|\left|1 + \frac{j\omega}{2}\right|} \angle \left[-90^\circ + \tan^{-1}\left(\frac{\omega}{10}\right) - \tan^{-1}\left(\frac{\omega}{2}\right) \right]$$

Compare the transfer function with the following expression:

$$\mathbf{H}(\omega) = H \angle \phi$$

The magnitude H is,

$$H = \frac{\left|1 + \frac{j\omega}{10}\right|}{|j\omega|\left|1 + \frac{j\omega}{2}\right|}$$

The phase ϕ is,

$$\phi = -90^\circ + \tan^{-1}\left(\frac{\omega}{10}\right) - \tan^{-1}\left(\frac{\omega}{2}\right) \dots\dots (1)$$

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Determine the magnitude H in decibels.

$$H_{dB} = 20 \log_{10} H$$

Substitute $\left(\frac{\left| 1 + \frac{j\omega}{10} \right|}{\left| j\omega \right| \left| 1 + \frac{j\omega}{2} \right|} \right)$ for H .

$$\begin{aligned} H_{dB} &= 20 \log_{10} \left(\frac{\left| 1 + \frac{j\omega}{10} \right|}{\left| j\omega \right| \left| 1 + \frac{j\omega}{2} \right|} \right) \\ &= 20 \log_{10} \left(\left| 1 + \frac{j\omega}{10} \right| \right) - 20 \log_{10} \left(\left| j\omega \right| \left| 1 + \frac{j\omega}{2} \right| \right) \\ &= 20 \log_{10} \left(\left| 1 + \frac{j\omega}{10} \right| \right) - 20 \log_{10} (|j\omega|) - 20 \log_{10} \left(\left| 1 + \frac{j\omega}{2} \right| \right) \\ H_{dB} &= 20 \log_{10} \left(\sqrt{1 + \left(\frac{\omega}{10} \right)^2} \right) - 20 \log_{10} (\omega) - 20 \log_{10} \left(\sqrt{1 + \left(\frac{\omega}{2} \right)^2} \right) \dots\dots (2) \end{aligned}$$

Step 3 of 9

The various terms of $H(\omega)$ are listed in the increasing order of their corner frequencies. Also the table 1 shows the slope contributed by each term and the change in slope at the corner frequency.

Table 1

Term	Corner Frequency (rad/sec)	Slope db/decade	Change in slope db/decade
$\frac{1}{j\omega}$	—	—20	—
$\frac{1}{1 + \frac{j\omega}{2}}$	$\omega = 2$	—20	$-20 - 20 = -40$
$1 + \frac{j\omega}{10}$	$\omega = 10$	+20	$-40 + 20 = -20$

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Choose a low frequency ω_l such that $\omega_l < \omega_{c1}$ and choose a high frequency ω_h such that $\omega_h > \omega_{c2}$.

Here, ω_{c1} is the first corner frequency whereas ω_{c2} is the highest corner frequency.

Let $\omega_l = 0.1$ rad/s and $\omega_h = 100$ rad/s

Calculate the magnitude at each frequency.

At $\omega = \omega_l$,

$$\begin{aligned} H_{dB} &= 20 \log \left| \frac{1}{\omega} \right| \\ &= 20 \log \left| \frac{1}{0.1} \right| \\ &= 20 \text{ dB} \end{aligned}$$

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At $\omega = \omega_{c1}$,

$$\begin{aligned} H_{dB} &= 20 \log \left| \frac{1}{\omega} \right| \\ &= 20 \log \left| \frac{1}{2} \right| \\ &= -6.02 \text{ dB} \end{aligned}$$

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At $\omega = \omega_{c2}$,

$$\begin{aligned} H_{dB} &= \left[\text{slope from } \omega_{c1} \text{ to } \omega_{c2} \times \log \frac{\omega_{c2}}{\omega_{c1}} \right] + A_{\omega=\omega_{c1}} \\ &= -40 \log \left(\frac{10}{2} \right) - 6.02 \\ &= -33.97 \text{ dB} \end{aligned}$$

At $\omega = \omega_h$,

$$\begin{aligned} H_{dB} &= \left[\text{slope from } \omega_{c2} \text{ to } \omega_h \times \log \frac{\omega_h}{\omega_{c2}} \right] + A_{\omega=\omega_{c2}} \\ &= -20 \log \left(\frac{100}{10} \right) - 33.97^\circ \\ &= -53.97 \text{ dB} \end{aligned}$$

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Tabulate the gain values according to the corner frequencies as shown in Table 2.

Table 2

ω	A
0.1	20 dB
2	-6.02 dB
10	-33.97 dB
100	-53.97 dB

Draw the magnitude plot as shown in figure 1.

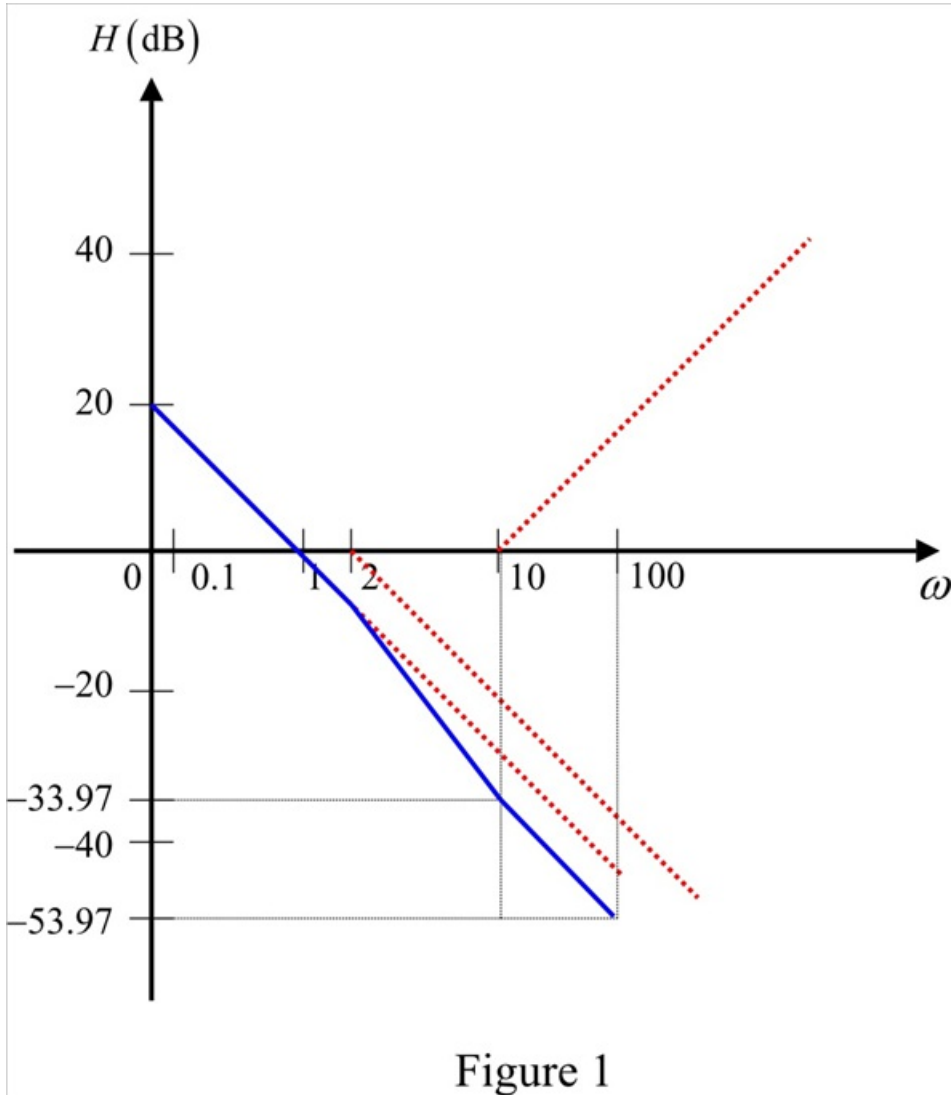


Figure 1

Step 8 of 9

The phase angle of ϕ are calculated for various values ω and listed in table 3

$$\phi = -90^\circ + \tan^{-1}\left(\frac{\omega}{10}\right) - \tan^{-1}\left(\frac{\omega}{2}\right)$$

Tabulate the values of ϕ for different corner frequencies as shown in Table 3.

Table 3

ω	ϕ
1	-110.85°
10	-123.69°
50	-99.02°
100	-94.56°

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Draw the phase plot as shown in figure 2.

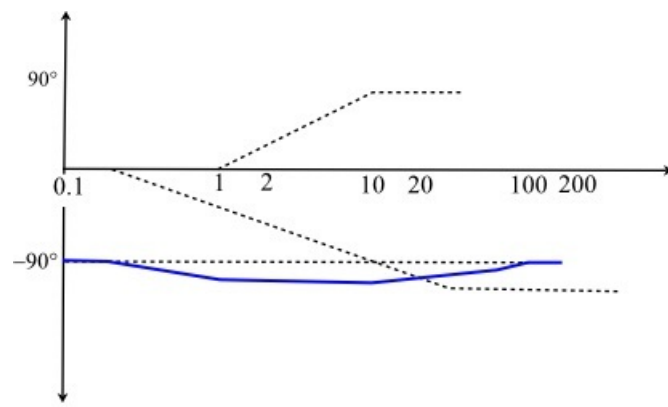


Figure 2